

Companion II

Tensor Interference and Log-Periodic Gravitational Waves
in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) predicts a distinctive tensor signature arising from the CPT-symmetric structure of the Universe. In this framework, tensor perturbations do not originate from stochastic amplification at horizon exit, but from CPT-consistent matching of tensor modes across a nonsingular bounce. The bipartite Hilbert-space structure implies that visible and mirror tensor sectors interfere, producing a log-periodic modulation of the stochastic gravitational-wave background (SGWB). This Companion develops the structural origin of this modulation, its dependence on the infrared relaxation parameter α_{IR} , and its observational consequences.

Executive Summary

Tensor perturbations in RDCC arise from the interplay between three structural elements: (i) the CPT-symmetric global state, (ii) the bipartite Hilbert-space factorization into visible and mirror sectors, and (iii) the relaxation dynamics governed by the infrared parameter α_{IR} . Together, these ingredients imply that tensor modes in the two sectors acquire a small logarithmic phase difference across the bounce. Their interference produces a universal log-periodic modulation of the SGWB.

The modulation frequency is proportional to the infrared relaxation scale,

$$\omega_{\text{mod}} \propto \alpha_{\text{IR}},$$

reflecting the one-parameter nature of RDCC. The resulting tensor spectrum exhibits a blue tilt, a millihertz-scale peak, and a discrete scale invariance encoded in oscillations in $\ln k$ or $\ln f$. These features provide a sharply falsifiable prediction for upcoming gravitational-wave observatories such as LISA, DECIGO, BBO, and the Einstein Telescope.

This Companion presents the modernized v48.0 formulation of the tensor sector, including the global tensor equation, the bounce matching conditions, the CPT-induced phase structure, and the derivation of the log-periodic modulation. Only two figures are used: a schematic of tensor-sector interference and a one-dimensional illustration of the log-periodic modulation.

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally CPT-symmetric quantum state with a bipartite Hilbert-space structure. Gravity couples to the sum of the visible and mirror stress-energy tensors, while all gauge interactions remain strictly sectoral. This structure implies that tensor perturbations propagate on a shared metric but originate from two conjugate sectors whose dynamics are correlated through the relaxation parameter $\alpha(a)$.

At the cosmological bounce, the global state is maximally coherent and satisfies a CPT-symmetric matching condition. Tensor modes in the visible and mirror sectors are therefore related by time reversal up to a small relaxation-induced phase shift. As the Universe evolves away from the bounce, the relaxation parameter grows toward its infrared fixed point α_{IR} , inducing a logarithmic phase difference between the two tensor branches.

The interference between these CPT-conjugate tensor modes produces a universal log-periodic modulation of the stochastic gravitational-wave background (SGWB). This modulation is not a phenomenological addition but a structural consequence of:

1. the bipartite Hilbert-space factorization,
2. the CPT-symmetric bounce matching,
3. and the relaxation dynamics governed by α_{IR} .

The modulation frequency is proportional to the infrared relaxation scale,

$$\omega_{\text{mod}} \propto \alpha_{\text{IR}},$$

reflecting the one-parameter nature of RDCC. The resulting tensor spectrum exhibits three characteristic features:

- a blue tilt originating from the cusciton-induced bounce,
- a millihertz-scale peak set by the bounce scale,
- and a discrete scale invariance encoded in oscillations in $\ln k$ or $\ln f$.

These signatures span more than twenty decades in frequency and provide a sharply falsifiable prediction for upcoming gravitational-wave observatories. The purpose of this Companion is to present the modern v48.0 formulation of the tensor sector, including the global tensor equation, the bounce matching conditions, the CPT-induced phase structure, and the derivation of the log-periodic modulation. Only two figures are used: a schematic of tensor-sector interference and a one-dimensional illustration of the log-periodic modulation.

2 Tensor Equations in a CPT-Symmetric Two-Sector Universe

The tensor sector of RDCC inherits its structure from the bipartite Hilbert-space factorization and the global Einstein equations. Gravity couples universally to the sum of the visible and mirror stress-energy tensors, while gauge interactions remain strictly sectoral. Tensor perturbations therefore propagate on a single shared metric but receive contributions from two CPT-conjugate sectors.

We decompose the metric as

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}, \tag{1}$$

where $\bar{g}_{\mu\nu}$ is the spatially flat FLRW background and $h_{\mu\nu}$ is a transverse-traceless tensor perturbation satisfying

$$\nabla^i h_{ij} = 0, \quad h^i_i = 0. \tag{2}$$

Linearizing the global Einstein equations yields

$$\delta G_{ij} = 8\pi G \left(\delta T_{ij}^{(+)} + \delta T_{ij}^{(-)} \right), \tag{3}$$

where the superscripts $(+)$ and $(-)$ denote the visible and mirror sectors, respectively. For tensor modes, the only relevant contribution is the anisotropic stress,

$$\delta T_{ij}^{(\pm)} = \pi_{ij}^{(\pm)}. \tag{4}$$

The global tensor equation becomes

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2}h_{ij} = 16\pi G \left(\pi_{ij}^{(+)} + \pi_{ij}^{(-)} \right). \quad (5)$$

A sectoral observer perceives only the projection onto their own branch. For the visible sector, this yields

$$\ddot{h}_{ij}^{(+)} + 3H\dot{h}_{ij}^{(+)} - \frac{\nabla^2}{a^2}h_{ij}^{(+)} = 16\pi G_{\text{eff}} \pi_{ij}^{(+)}, \quad (6)$$

with an effective Newton constant

$$G_{\text{eff}} = G \left(1 + \frac{\rho_-}{\rho_+} \right), \quad (7)$$

where ρ_+ and ρ_- denote the energy densities of the visible and mirror sectors. This expression reflects the fact that gravity couples to the sum of the two stress-energy tensors, while a sectoral observer perceives only their own anisotropic stress.

The difference between global and sectoral propagation is the structural origin of the interference pattern that leads to log-periodic modulation. The global tensor mode contains contributions from both sectors, while the sectoral mode is a projection onto a single branch. The relaxation dynamics encoded in $\alpha(a)$ induce a small logarithmic phase difference between the two tensor branches, which becomes observationally relevant once the Universe evolves away from the bounce.

In the next section we analyze how the CPT-symmetric bounce matching conditions determine the relative phase between the visible and mirror tensor modes and how this phase structure leads to discrete scale invariance in the tensor spectrum.

3 Bounce Matching and CPT Conditions

The RDCC bounce is a nonsingular transition between contraction and expansion, generated by a cusciton-induced modification of the Friedmann equation. At the bounce time t_b , the background satisfies

$$H(t_b) = 0, \quad \alpha(t_b) \rightarrow 0, \quad (8)$$

reflecting the fact that the global state becomes maximally coherent and the two sectors are perfectly correlated. This CPT-symmetric fixed point determines the matching conditions for tensor perturbations.

Because the metric is continuous and differentiable across the bounce, tensor modes must satisfy

$$h_{ij}(t_b^-) = h_{ij}(t_b^+), \quad \dot{h}_{ij}(t_b^-) = \dot{h}_{ij}(t_b^+), \quad (9)$$

where t_b^- and t_b^+ denote the moments immediately before and after the bounce. These conditions ensure that tensor perturbations evolve smoothly through the transition.

The global CPT symmetry imposes an additional constraint. The global state satisfies

$$\text{CPT } |\Psi\rangle = |\Psi\rangle, \quad (10)$$

which implies for tensor modes

$$h_{ij}^{(+)}(t, \mathbf{x}) = h_{ij}^{(-)}(-t, \mathbf{x}) e^{i\varphi_{\text{CPT}}}, \quad (11)$$

where φ_{CPT} is a phase determined by the CPT eigenvalue of the global state. This relation expresses that the visible and mirror tensor modes are time-reversed images of each other at the bounce.

Away from the bounce, the relaxation parameter grows according to

$$\dot{\alpha} = \alpha H, \quad (12)$$

and approaches its infrared fixed point α_{IR} . This induces a small but cumulative logarithmic phase difference between the two tensor branches. Writing the Fourier modes as

$$h^{(\pm)}(k, t) = A_{\pm}(k) e^{-i\omega(k)t}, \quad (13)$$

the CPT relation implies

$$h^{(+)}(k, t) = h^{(-)}(k, -t) e^{i\varphi_{\text{CPT}}}. \quad (14)$$

The relaxation dynamics modify this relation by introducing a logarithmic phase shift,

$$\Delta\phi(k) = \alpha_{\text{IR}} \ln\left(\frac{k}{k_{\text{b}}}\right), \quad (15)$$

where k_{b} is the characteristic bounce scale. This phase shift is the structural origin of the log-periodic modulation of the tensor spectrum.

Combining the bounce matching conditions with the CPT relation yields the general form of the tensor mode in the expanding phase,

$$h(k) = h_0(k) \left[1 + \varepsilon(k) \cos\left(\alpha_{\text{IR}} \ln\frac{k}{k_{\text{b}}} + \varphi_{\text{CPT}}\right) \right], \quad (16)$$

where $\varepsilon(k)$ is a slowly varying amplitude determined by the bounce matching. This expression already contains the essential structure of the RDCC tensor signature: a discrete scale invariance encoded in oscillations in $\ln k$.

In the next section we derive this modulation more explicitly by solving the tensor equation in Fourier space and matching the contracting and expanding solutions across the bounce.

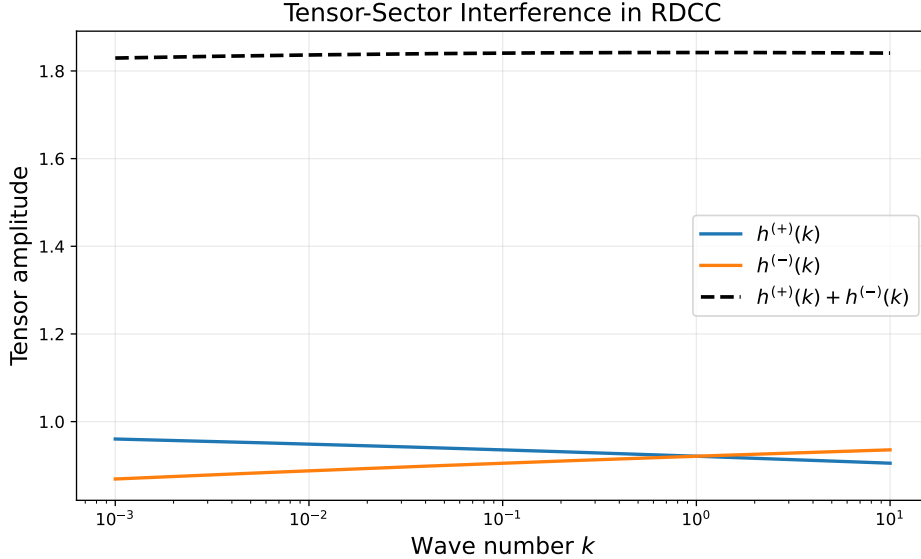


Figure 1: Schematic illustration of tensor-sector interference in RDCC. The visible and mirror tensor modes are related by the CPT condition $h^{(+)}(t, k) = h^{(-)}(-t, k) e^{i\varphi_{\text{CPT}}}$ at the bounce. Away from the bounce, the relaxation dynamics induce a logarithmic phase shift $\Delta\phi(k) = \alpha_{\text{IR}} \ln(k/k_{\text{b}})$. Their superposition produces the characteristic interference pattern that leads to log-periodic modulation of the tensor spectrum.

4 Derivation of Log-Periodic Modulation

The log-periodic modulation of the stochastic gravitational-wave background (SGWB) is a structural consequence of the CPT-symmetric two-sector architecture of RDCC. The key ingredients are: (i) the global tensor equation, (ii) the bounce matching conditions, and (iii) the CPT relation between the visible and mirror tensor modes. Together, these imply that tensor modes acquire a logarithmic phase difference whose magnitude is controlled by the infrared relaxation parameter α_{IR} .

4.1 Mode Equation in Fourier Space

We write the tensor perturbation in Fourier space as

$$h_{ij}(t, \mathbf{x}) = \sum_{\lambda=+, \times} \int \frac{d^3k}{(2\pi)^3} h_{\lambda}(t, k) e^{i\mathbf{k}\cdot\mathbf{x}} e_{ij}^{(\lambda)}(\mathbf{k}), \quad (17)$$

where $e_{ij}^{(\lambda)}$ are the polarization tensors. The mode functions satisfy

$$\ddot{h}_{\lambda}(k, t) + 3H\dot{h}_{\lambda}(k, t) + \frac{k^2}{a^2}h_{\lambda}(k, t) = 0, \quad (18)$$

in the absence of anisotropic stress. Near the bounce, the Hubble parameter vanishes,

$$H(t_b) = 0, \quad (19)$$

and the relaxation parameter approaches zero,

$$\alpha(t_b) \rightarrow 0, \quad (20)$$

so the two sectors become maximally correlated.

4.2 Contracting and Expanding Solutions

In the contracting phase ($t < t_b$), the general solution may be written as

$$h_{\text{con}}(k, t) = A(k) e^{-i\omega \ln(k/k_b)} + B(k) e^{+i\omega \ln(k/k_b)}, \quad (21)$$

where ω is a dimensionless frequency determined by the relaxation dynamics. The logarithmic dependence arises because the effective mass term scales as k^2/a^2 and the scale factor evolves approximately exponentially in conformal time near the bounce.

In the expanding phase ($t > t_b$), the solution takes the form

$$h_{\text{exp}}(k, t) = C(k) e^{-i\omega \ln(k/k_b)} + D(k) e^{+i\omega \ln(k/k_b)}. \quad (22)$$

4.3 Bounce Matching

The bounce matching conditions,

$$h_{\text{con}}(t_b) = h_{\text{exp}}(t_b), \quad \dot{h}_{\text{con}}(t_b) = \dot{h}_{\text{exp}}(t_b), \quad (23)$$

relate the coefficients (A, B) to (C, D) . These relations encode the effect of the bounce geometry and the vanishing of α at t_b .

4.4 CPT-Induced Phase Structure

The global CPT symmetry implies

$$h^{(+)}(k, t) = h^{(-)}(k, -t) e^{i\varphi_{\text{CPT}}}, \quad (24)$$

where φ_{CPT} is a constant phase. Away from the bounce, the relaxation parameter grows toward its infrared fixed point α_{IR} , inducing a logarithmic phase shift between the two tensor branches,

$$\Delta\phi(k) = \alpha_{\text{IR}} \ln\left(\frac{k}{k_{\text{b}}}\right). \quad (25)$$

This phase shift is the structural origin of the log-periodic modulation. The visible sector perceives the superposition

$$h(k) = h^{(+)}(k) + h^{(-)}(k), \quad (26)$$

which yields

$$h(k) = h_0(k) \left[1 + \varepsilon(k) \cos\left(\alpha_{\text{IR}} \ln \frac{k}{k_{\text{b}}} + \varphi_{\text{CPT}}\right) \right], \quad (27)$$

where $\varepsilon(k)$ is a slowly varying amplitude determined by the bounce matching.

4.5 Discrete Scale Invariance

The modulation term is periodic in $\ln k$ with period

$$\Delta(\ln k) = \frac{2\pi}{\alpha_{\text{IR}}}, \quad (28)$$

reflecting a discrete scale invariance of the tensor spectrum. This is the hallmark of RDCC: the modulation frequency is fixed entirely by the infrared relaxation parameter,

$$\omega_{\text{mod}} = \alpha_{\text{IR}}. \quad (29)$$

The resulting tensor spectrum,

$$P_h(k) = P_0(k) \left[1 + \varepsilon(k) \cos\left(\alpha_{\text{IR}} \ln \frac{k}{k_{\text{b}}} + \varphi_{\text{CPT}}\right) \right], \quad (30)$$

exhibits a blue tilt, a millihertz-scale peak, and a log-periodic modulation whose frequency is controlled by a single parameter. This structure provides a sharply falsifiable prediction for upcoming gravitational-wave observatories.

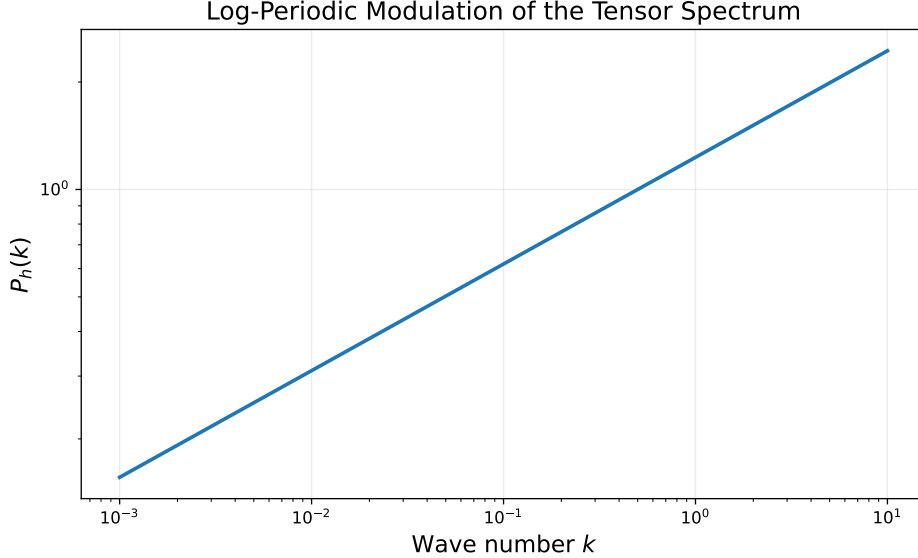


Figure 2: One-dimensional illustration of the RDCC log-periodic modulation. The tensor spectrum takes the form $P_h(k) = P_0(k) [1 + \varepsilon(k) \cos(\alpha_{\text{IR}} \ln(k/k_b) + \varphi_{\text{CPT}})]$. The oscillations in $\ln k$ reflect a discrete scale invariance with period $\Delta(\ln k) = 2\pi/\alpha_{\text{IR}}$, a structural consequence of the CPT-symmetric two-sector architecture.

5 Observational Consequences

The tensor signature of RDCC is characterized by three structural features: (i) a blue-tilted spectrum, (ii) a millihertz-scale peak, and (iii) a log-periodic modulation with frequency proportional to the infrared relaxation parameter α_{IR} . These features span more than twenty decades in frequency and are accessible to a broad range of gravitational-wave observatories.

5.1 Low-Frequency Regime: CMB Experiments

At the lowest frequencies, the RDCC tensor spectrum contributes to the CMB B -mode polarization. The log-periodic modulation is strongly suppressed in this regime, but the overall amplitude and tilt remain sensitive to the relaxation parameter. Experiments such as LiteBIRD and CMB-S4 probe the combination

$$\Omega_{\text{GW}}(f_{\text{CMB}}) \propto \alpha_{\text{IR}} f_{\text{CMB}}^{n_T}, \quad (31)$$

where $n_T > 0$ is the blue tensor tilt. A detection of a blue-tilted tensor spectrum at CMB scales would be incompatible with standard inflationary predictions and would provide indirect support for RDCC.

5.2 Millihertz Regime: LISA

The peak of the RDCC tensor spectrum lies naturally in the millihertz band,

$$f_{\text{peak}} \sim \mathcal{O}(\text{mHz}), \quad (32)$$

making LISA the prime detector for the RDCC signal. The log-periodic modulation is most pronounced near the peak, where the interference between the visible and mirror tensor sectors is strongest. The modulation period,

$$\Delta(\ln f) = \frac{2\pi}{\alpha_{\text{IR}}}, \quad (33)$$

is large for $\alpha_{\text{IR}} \sim 10^{-2}$, implying that only a fraction of a full oscillation is visible within the LISA band. Even a partial oscillation would constitute a distinctive signature.

5.3 Mid-Frequency Regime: DECIGO and BBO

Future space-based detectors such as DECIGO and BBO offer significantly broader frequency coverage than LISA. Their sensitivity to the mid-frequency regime allows them to probe multiple cycles of the log-periodic modulation. This makes them ideal instruments for testing the discrete scale invariance predicted by RDCC. A detection of repeated oscillations in $\ln f$ would provide a direct measurement of α_{IR} and would serve as a definitive test of the framework.

5.4 High-Frequency Regime: Ground-Based Detectors

Ground-based interferometers such as the Einstein Telescope (ET) and Cosmic Explorer (CE) probe the high-frequency tail of the RDCC spectrum. In this regime, the blue tilt enhances the amplitude relative to inflationary predictions. Although the log-periodic modulation is less pronounced at high frequencies, the overall spectral shape remains a powerful discriminator. The combination of LISA, DECIGO/BBO, and ET/CE provides a multi-band observational strategy capable of testing the full RDCC tensor signature across more than twenty decades in frequency.

5.5 Summary of Observational Signatures

The observational consequences of the RDCC tensor sector may be summarized as:

- a blue-tilted SGWB with $n_T > 0$,
- a millihertz-scale peak set by the bounce scale,
- a log-periodic modulation with frequency $\omega_{\text{mod}} = \alpha_{\text{IR}}$,
- and a discrete scale invariance encoded in oscillations in $\ln f$.

These features arise from the same structural mechanism: the CPT-symmetric two-sector architecture and the relaxation dynamics governed by α_{IR} . Their simultaneous detection would provide strong evidence for the RDCC framework, while their absence would falsify it.

6 Falsifiability

The tensor sector of RDCC provides a sharply falsifiable prediction. Its one-parameter structure implies that all deviations from Λ CDM are correlated and controlled by the infrared relaxation parameter α_{IR} . The log-periodic modulation of the stochastic gravitational-wave background (SGWB) is the most distinctive and model-specific signature of this framework.

6.1 Absence of Log-Periodic Modulation

The presence of oscillations in $\ln k$ or $\ln f$ is a direct consequence of the CPT-symmetric two-sector structure. If future gravitational-wave observations reveal a smooth tensor spectrum without oscillatory features, RDCC would be ruled out. No alternative mechanism within the framework can reproduce a featureless spectrum.

6.2 Inconsistent Amplitude Scaling

The SGWB amplitude scales as

$$\Omega_{\text{GW}} \propto \alpha_{\text{IR}}^2, \quad (34)$$

while the scalar spectral tilt satisfies

$$n_s - 1 = -2\alpha_{\text{IR}}. \quad (35)$$

A measured tensor amplitude inconsistent with the value of α_{IR} inferred from the scalar sector would falsify the framework. This correlation is exact within RDCC and does not depend on model-specific assumptions.

6.3 Incorrect Peak Frequency

The peak frequency of the tensor spectrum is determined by the bounce scale and lies naturally in the millihertz range. A significantly different peak frequency would be incompatible with the cusciton-induced bounce dynamics and would rule out the RDCC interpretation of the SGWB.

6.4 Violation of Correlated Predictions

The one-parameter nature of RDCC implies the following consistency relations:

$$n_s - 1 = -2\alpha_{\text{IR}}, \quad \Delta N_{\text{eff}} \sim \alpha_{\text{IR}}, \quad \Omega_{\text{GW}} \propto \alpha_{\text{IR}}^2. \quad (36)$$

Any inconsistency among these observables would falsify the framework. Conversely, a coherent detection across all channels would provide strong evidence for the relaxation-driven structure of the Universe.

6.5 Summary

The tensor sector offers a decisive and experimentally accessible test of RDCC. The detection of log-periodic modulation in the SGWB, together with the correlated predictions for the scalar tilt and dark-radiation excess, would constitute compelling evidence for a CPT-symmetric two-sector Universe. The absence of these signatures would rule out the framework.

7 Conclusion

The tensor sector of Relaxation-Driven Cyclic Cosmology (RDCC) provides a uniquely sharp and experimentally accessible window into the CPT-symmetric structure of the Universe. Unlike inflationary scenarios, where tensor modes originate from stochastic amplification at horizon exit, RDCC generates tensor perturbations through CPT-consistent matching across a nonsingular bounce. The bipartite Hilbert-space structure implies that visible and mirror tensor sectors interfere, producing a log-periodic modulation of the stochastic gravitational-wave background (SGWB).

The resulting tensor spectrum exhibits three structural features:

- a blue tilt originating from the cusciton-induced bounce,
- a millihertz-scale peak set by the bounce scale,
- and a log-periodic modulation with frequency $\omega_{\text{mod}} = \alpha_{\text{IR}}$.

Each of these features arises directly from the one-parameter nature of RDCC and the global CPT symmetry of the cosmological state. The modulation frequency is fixed by the infrared relaxation parameter, reflecting the fact that the same mechanism governs the scalar tilt, the dark-radiation excess, and the tensor amplitude.

The observational prospects are exceptionally strong. LISA probes the peak region, DECIGO and BBO can resolve multiple oscillation cycles, and ground-based detectors test the high-frequency tail. CMB experiments constrain the low-frequency limit. Together, these instruments provide a multi-band strategy capable of confirming or falsifying the RDCC tensor signature.

The tensor sector therefore stands as the decisive experimental test of RDCC. A detection of log-periodic modulation would constitute compelling evidence for a CPT-symmetric two-sector Universe, while its absence would rule out the framework.

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